

Describe the type(s) of research subjects or activities that interest you at your first and second choice host laboratories, and discuss any particular factors influencing your choice of host laboratories.

I am interested in research that models complex, large-scale physical systems using molecular dynamics and quantum algorithms. I'm seeking environments where I can deepen my understanding of large-scale simulation, hybrid quantum-classical computing, and optimization methods to solve scientific problems in material science, chemistry, and biology. Oak Ridge National Laboratory (ORNL) and Brookhaven National Laboratory (BNL) strongly align with these goals.

At ORNL, I aim to build on my experience with hybrid quantum-classical workflows and the molecular dynamics of biomolecular systems. My previous work involved group theoretic methods to minimize the resources needed for the classical simulation of quantum circuits, and I'm interested in learning more about other hybrid quantum-classical approaches for applications in chemistry, material science and combinatorial optimization. I am particularly excited about the Quantum Computational Science Group under Dr. Ryan Bennink and their Quantum High Performance Computing (Q-HPC) initiative, which focuses on integrating quantum algorithms with exascale classical computing. My background in quantum algorithms, high performance computing, molecular simulation, and Monte Carlo optimization provide a strong foundation to contribute to this initiative.

My work as part of the MSU-DOE Plant Research Laboratory has often involved modelling large biomolecular systems. These are expensive to simulate for longer periods of time, which limits the range of biological phenomena that can be studied. Through the Molecular Biophysics Group and Oak Ridge Leadership Computing Facility at ORNL, I hope to learn how large-scale molecular dynamics and deep learning based structure prediction methods can give new insights into biomolecular processes.

At BNL, I'm drawn towards the Computing and Data Sciences Directorate and their focus on quantum algorithms for quantum chromodynamics simulation. My previous experience has involved exploiting physical structure to reduce the resource requirements of quantum computation, and I am eager to learn how such optimizations can be applied in high energy physics. I am also interested in contributing to exascale numerical algorithms for solving problems in material science and biology as part of the Computational Science Initiative at BNL. By doing so, I hope to contribute to HPC-workflow engineering and quantum circuit design.

In choosing my host laboratories, I prioritized settings where my existing strengths would be directly applicable to ongoing research, while also providing me with unique opportunities. ORNL's exascale computing resources and Q-HPC initiative, as well as BNL's commitment to exascale numerical algorithms and physics-informed quantum algorithms fit my trajectory towards pursuing graduate studies in computational science.